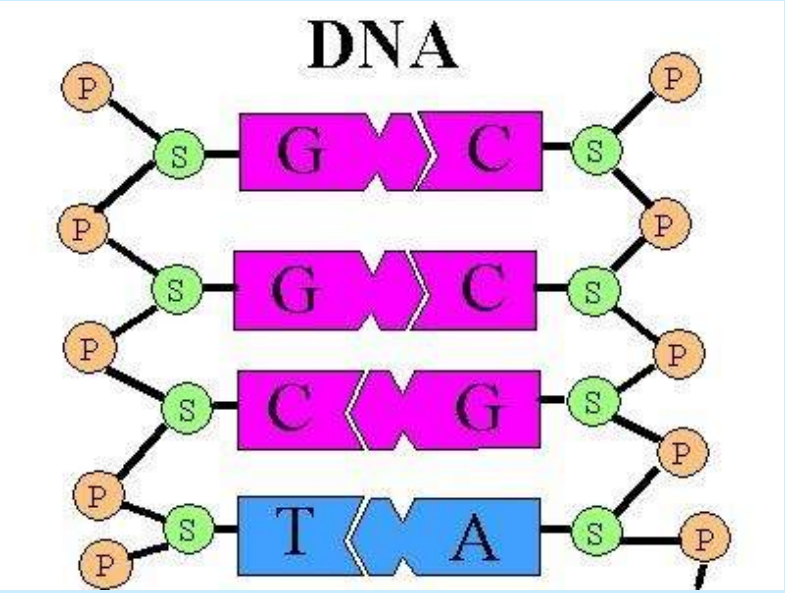
DNA Replication

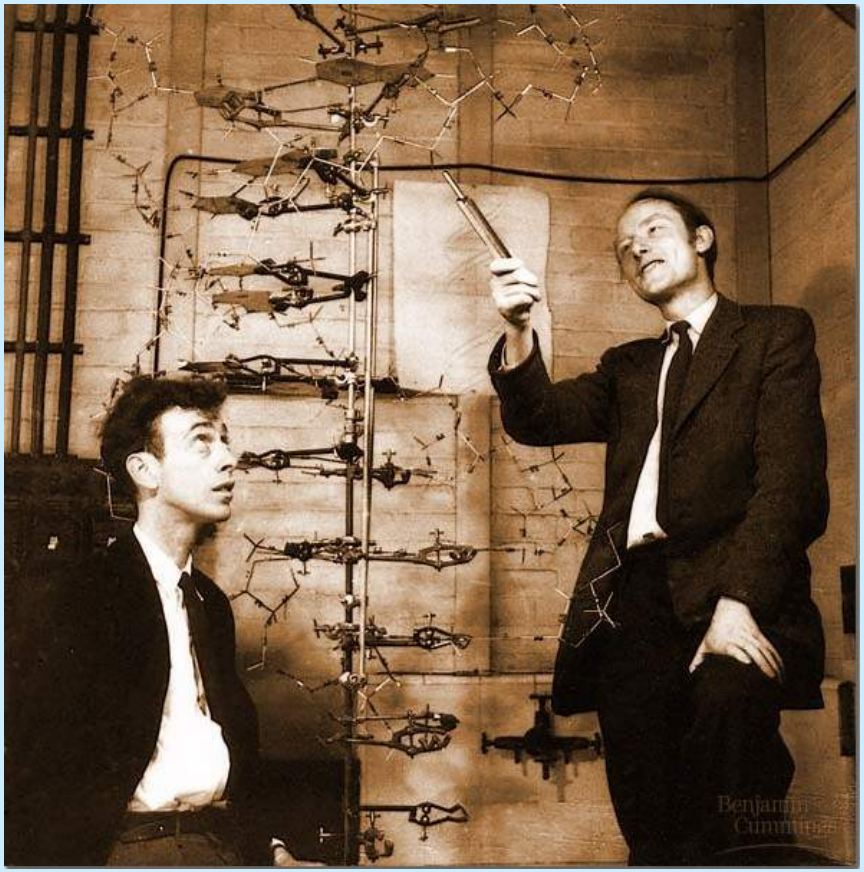
Discovering the structure of DNA

- DNA = **D**eoxyribose **n**ucleic **a**cid
- Made out of sugars (deoxyribose), phosphates and nitrogen bases



Discovering the structure of DNA

- Structure was discovered in 1953 by James Watson and Francis Crick



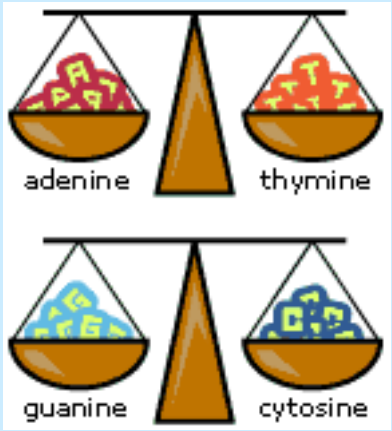
Discovering the structure of DNA

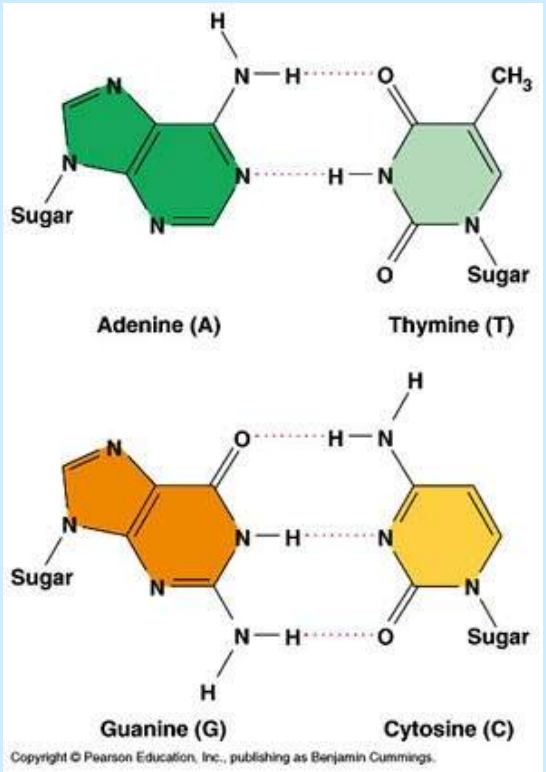
Rosalind Franklin's DNA image



"Chargoff's rule"

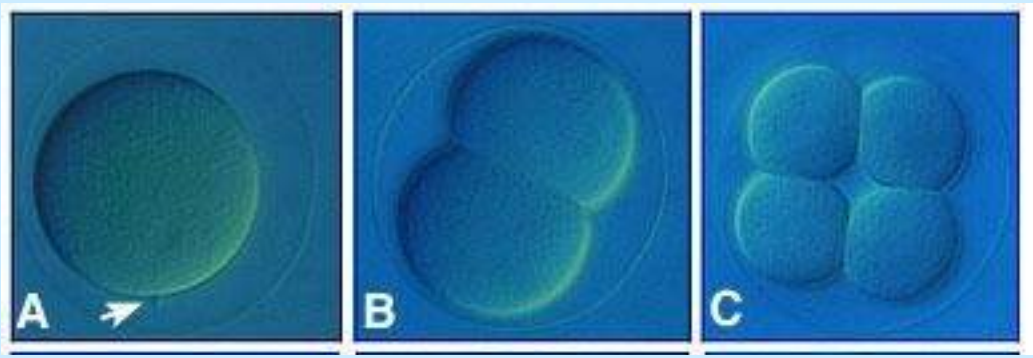
A = T & G = C





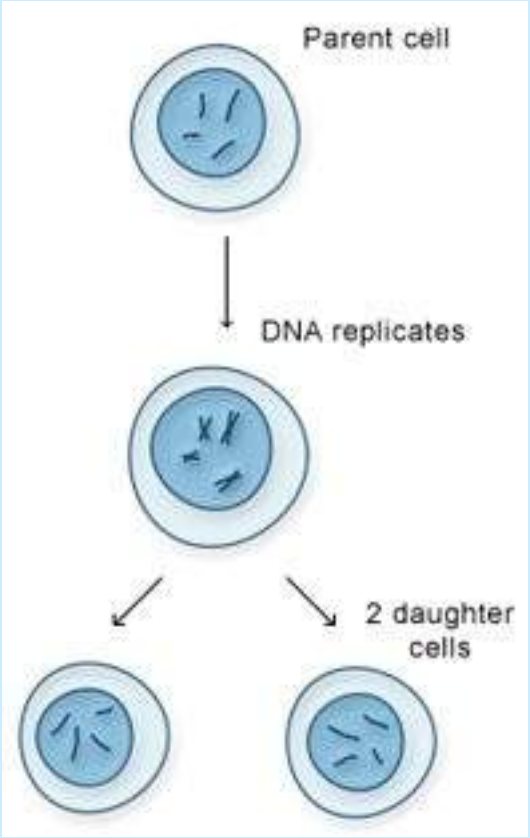
Cell division and DNA replication

- Cells divide
  - Growth, Repair, Replacement



- Before cells divide they have to double cell structures, organelles and their genetic information

DNA replication



DNA stores information in the sequence of its bases

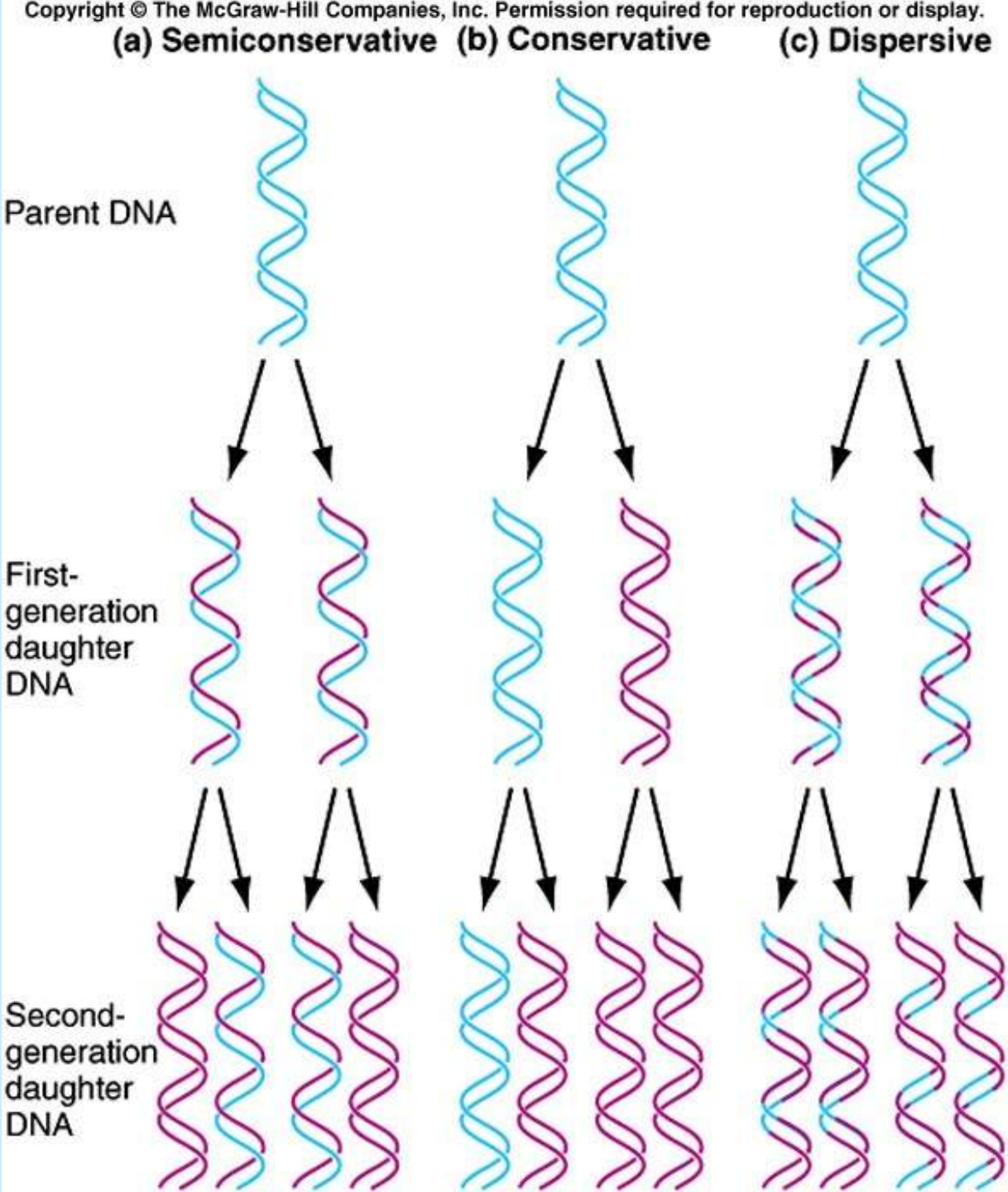
- **Much of DNA's sequence-specific information is accessible only when the double helix is unwound**
- **Proteins read the DNA sequence of nucleotides as the DNA helix unwinds.**
- **Proteins can either bind to a DNA sequence, or initiate the copying of it.**
- **Some genetic information is accessible even in intact, double-stranded DNA molecules**
- **Some proteins recognize the base sequence of DNA without unwinding it (One example is a restriction enzyme).**

Basic rules of replication

A. Semi-conservative
B. Starts at the ‘origin’
C. Can be uni or bidirectional
D. Semi-discontinuous
E. Synthesis always in the 5-3′ direction
F. RNA primers required

DNA replication

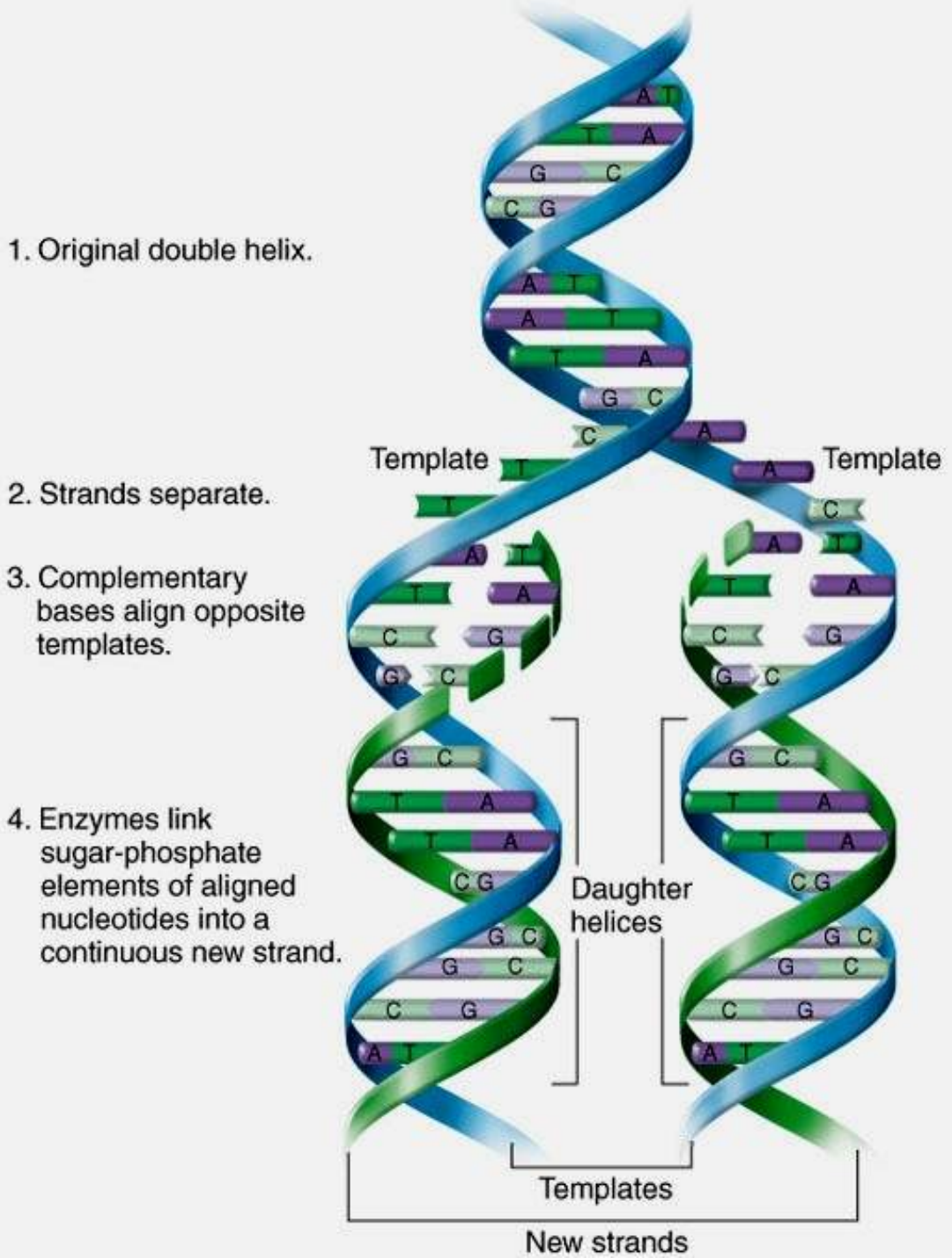
3 possible models



Semi-conservative replication:

One strand of duplex passed on unchanged to each of the daughter cells.

This 'conserved' strand acts as a template for the synthesis of a new, complementary strand by the enzyme DNA polymerase



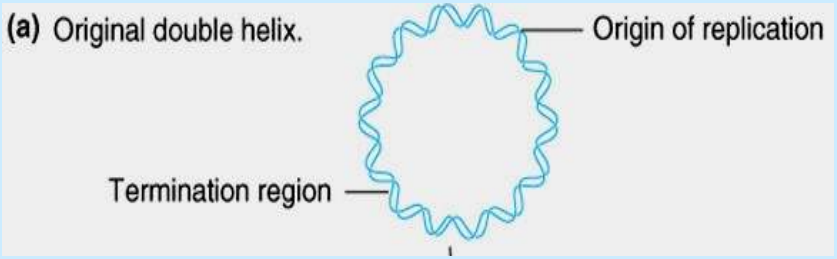
B) Starts at origin

Initiator proteins identify specific base sequences on DNA called sites of origin

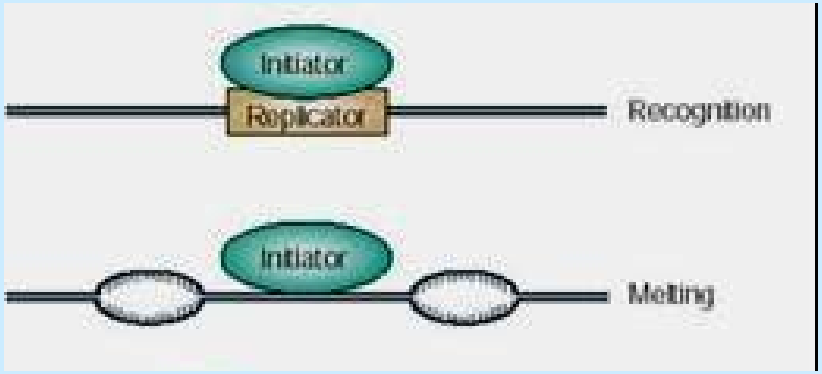
Eukaryotes – multiple sites of origin (replicator)

Prokaryotes – single origin site E.g *E.coli* - *oriC*

E.g. yeast - ARS (autonomously replicating sequences)



Prokaryotes

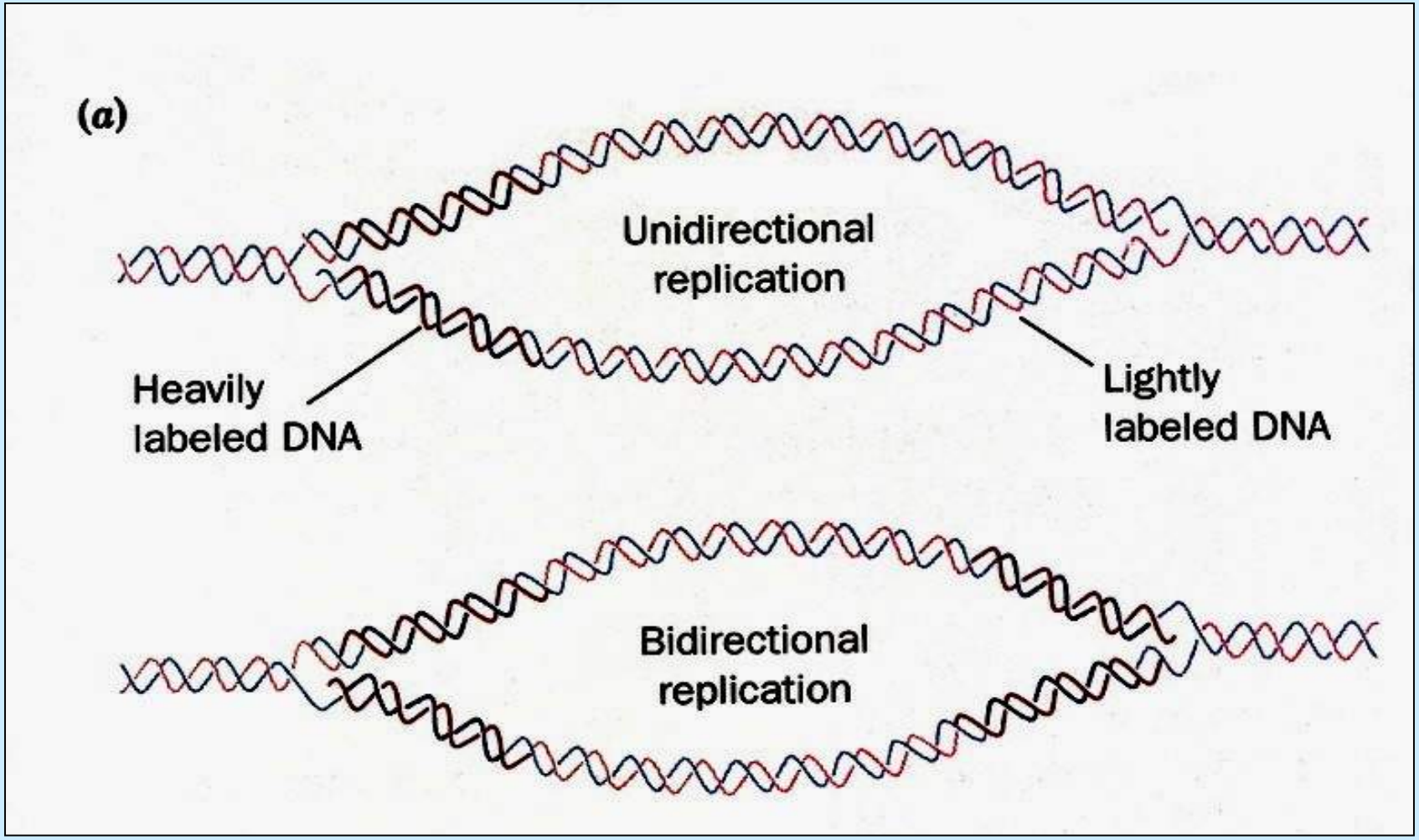


Eukaryotes

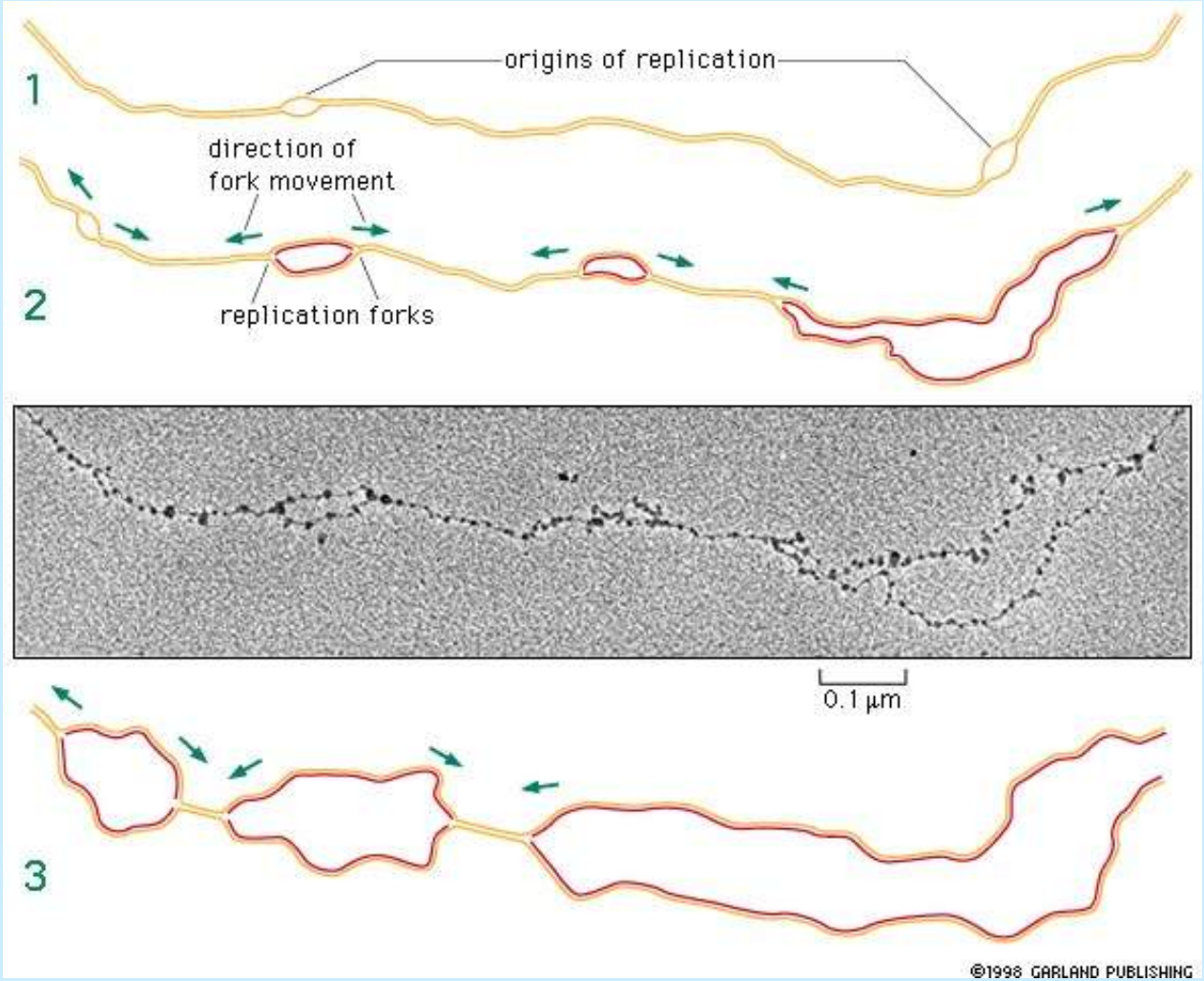
In eukaryotes, replication only occurs during the S phase of the cell cycle.

D) Uni or bidirectional

- Replication forks move in one or opposite directions



Replication Fork

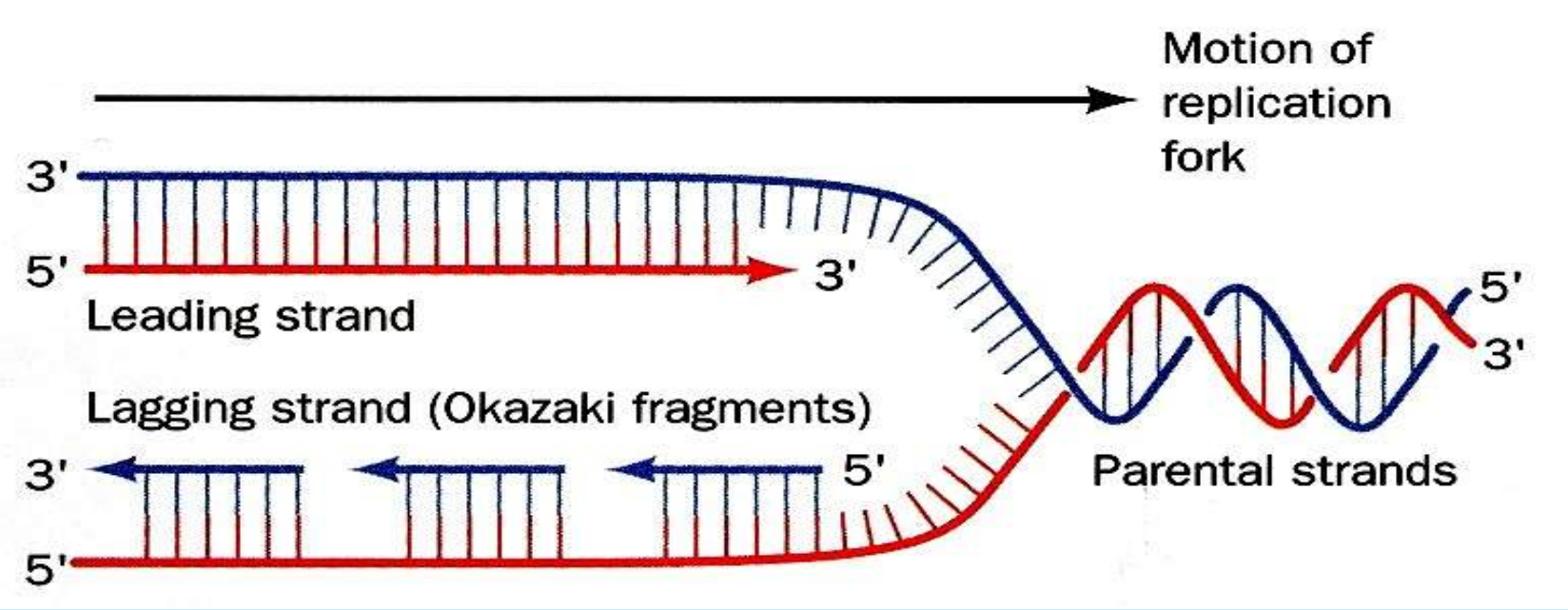


- Bidirectional movement of the DNA replication machinery

E) Semi-discontinuous replication

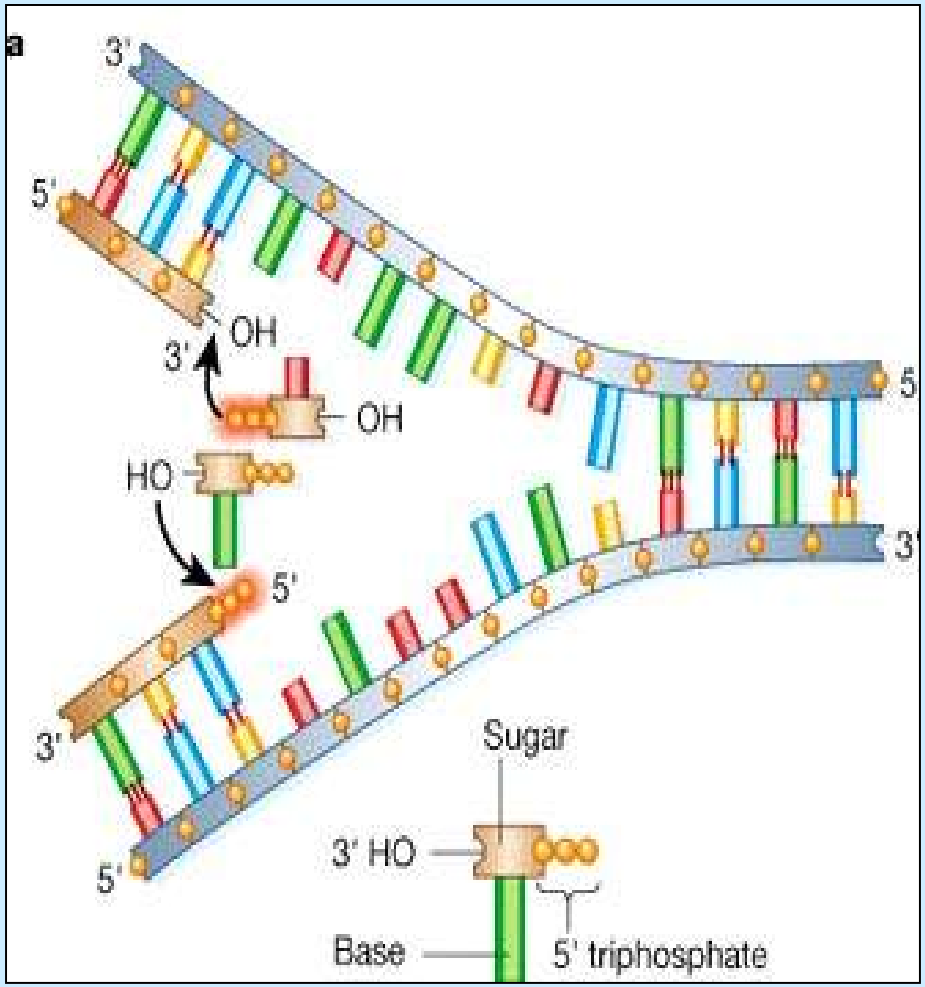
Anti parallel strands replicated simultaneously

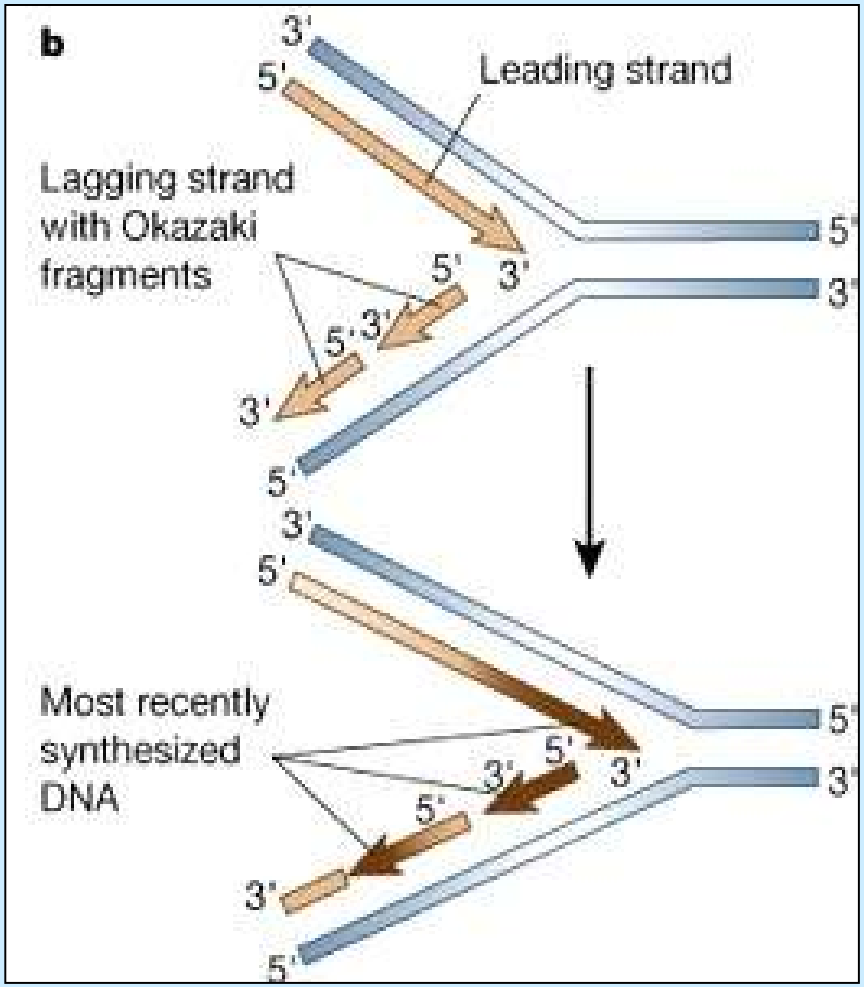
- Leading strand synthesis **continuously** in 5′– 3′
- Lagging strand synthesis in **fragments** in 5′-3′



Semi-discontinuous replication

New strand synthesis always in the 5′-3′ direction





Core proteins at the replication fork

Topoisomerases	**- Prevents torsion by DNA breaks**
Helicases	**- separates 2 strands**
Primase	**- RNA primer synthesis**
Single strand binding proteins	**- prevent reannealing of single strands**
DNA polymerase	**- synthesis of new strand**
Tethering protein	**- stabilises polymerase**
DNA ligase	**- seals nick via phosphodiester linkage**

MECHANISM

OF

DNA REPLICATION

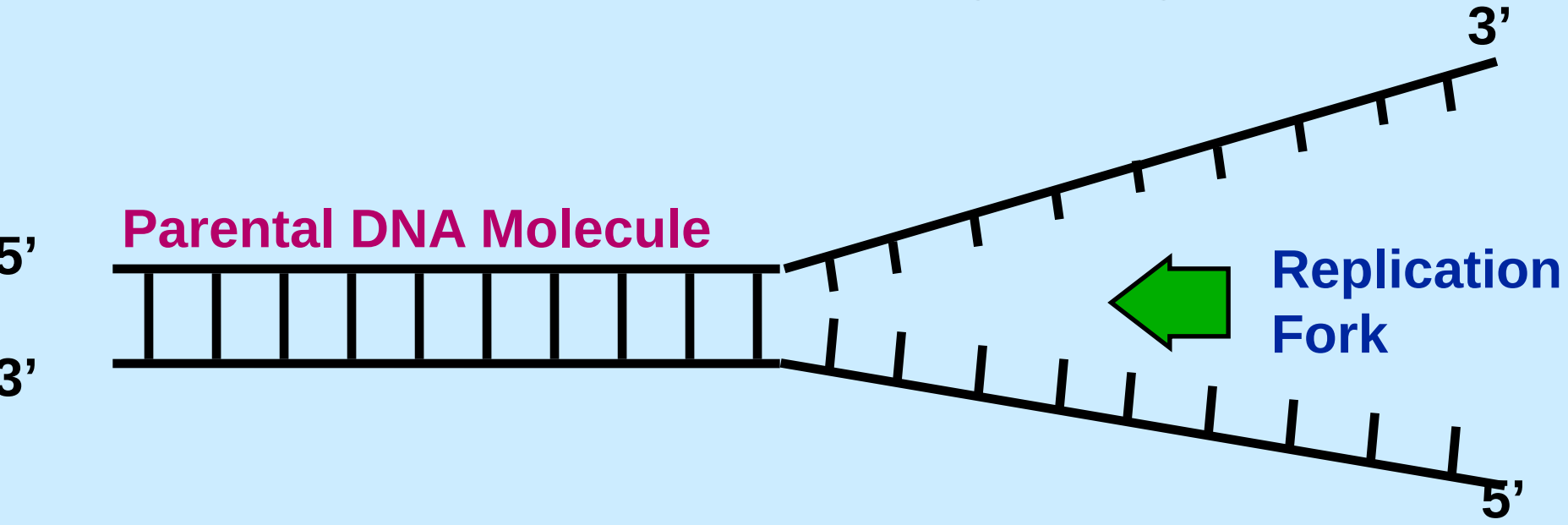
The mechanism of DNA replication

Arthur Kornberg, a Nobel prize winner and other biochemists deduced steps of replication

- Initiation
  - Proteins bind to DNA and open up double helix
  - Prepare DNA for complementary base pairing
- Elongation
  - Proteins connect the correct sequences of nucleotides into a continuous new strand of DNA
- Termination
  - Proteins release the replication complex

DNA Replication

- **Origins of replication**
  1. **Replication Forks**: **hundreds** of **Y-shaped** regions of **replicating DNA molecules** where new strands are growing.



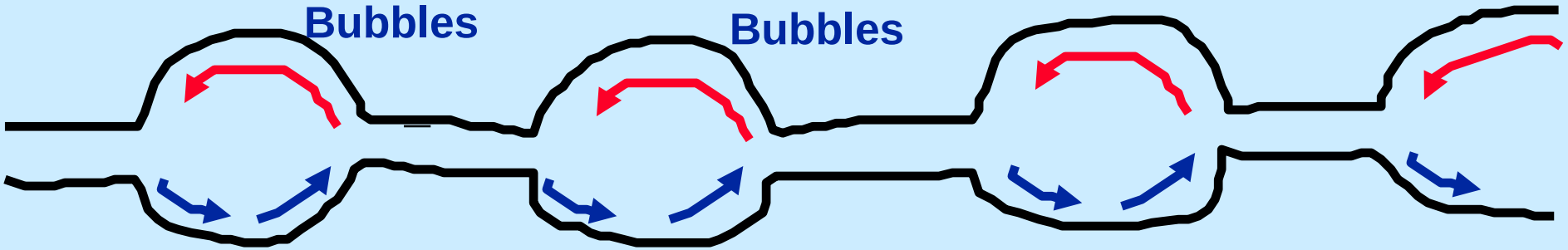
DNA Replication

- **Origins of replication**

  2. Replication Bubbles:

  a. Hundreds of replicating bubbles (Eukaryotes).

  b. Single replication fork (bacteria).



DNA Replication

- **Strand Separation**:

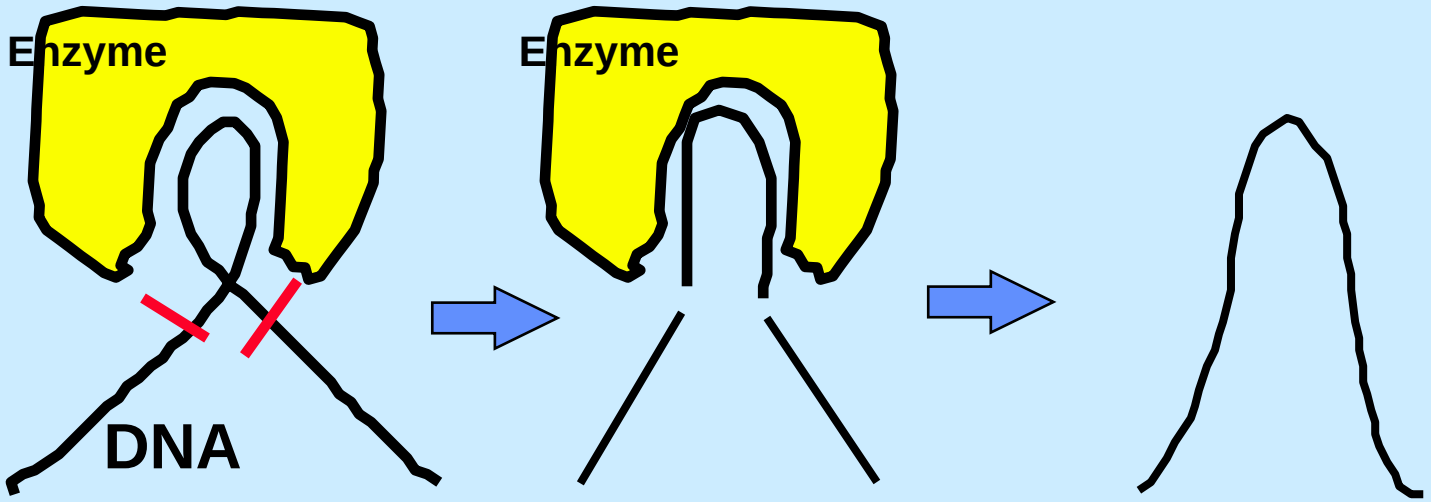
  1. **Helicase**: enzyme which catalyze the **unwinding** and **separation** (breaking H-Bonds) of the parental double helix.

  2. **Single-Strand Binding Proteins**: proteins which attach and help keep the separated strands apart.

DNA Replication

- **Strand Separation**:

  3. **Topoisomerase**: **enzyme which relieves stress** on the **DNA molecule** by allowing free rotation around a single strand.



DNA Replication

- **Priming:**

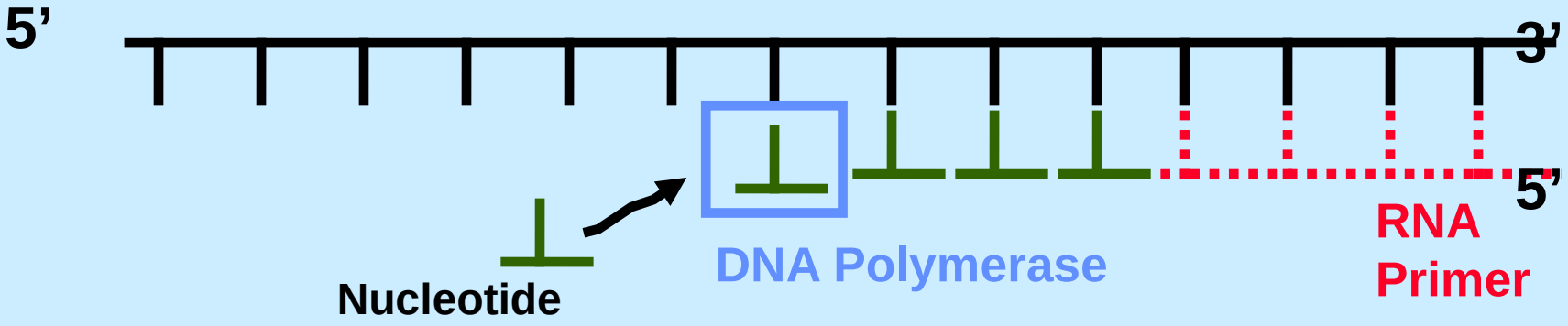
  1. **RNA primers**: before new DNA strands can form, there must be small pre-existing **primers (RNA)** present to start the addition of new nucleotides **(DNA Polymerase)**.

  2. **Primase**: enzyme that polymerizes (synthesizes) the **RNA Primer**.

DNA Replication

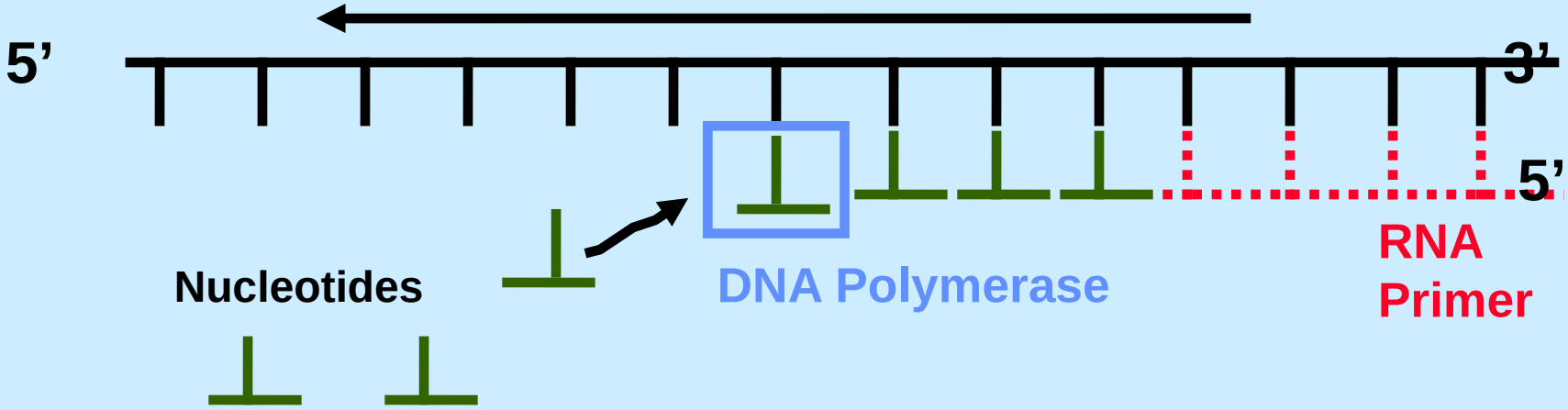
- **Synthesis of the new DNA Strands:**

1. **DNA Polymerase**: with a **RNA primer** in place, DNA Polymerase (enzyme) catalyze the **synthesis of a new DNA strand in the 5' to 3' direction**.



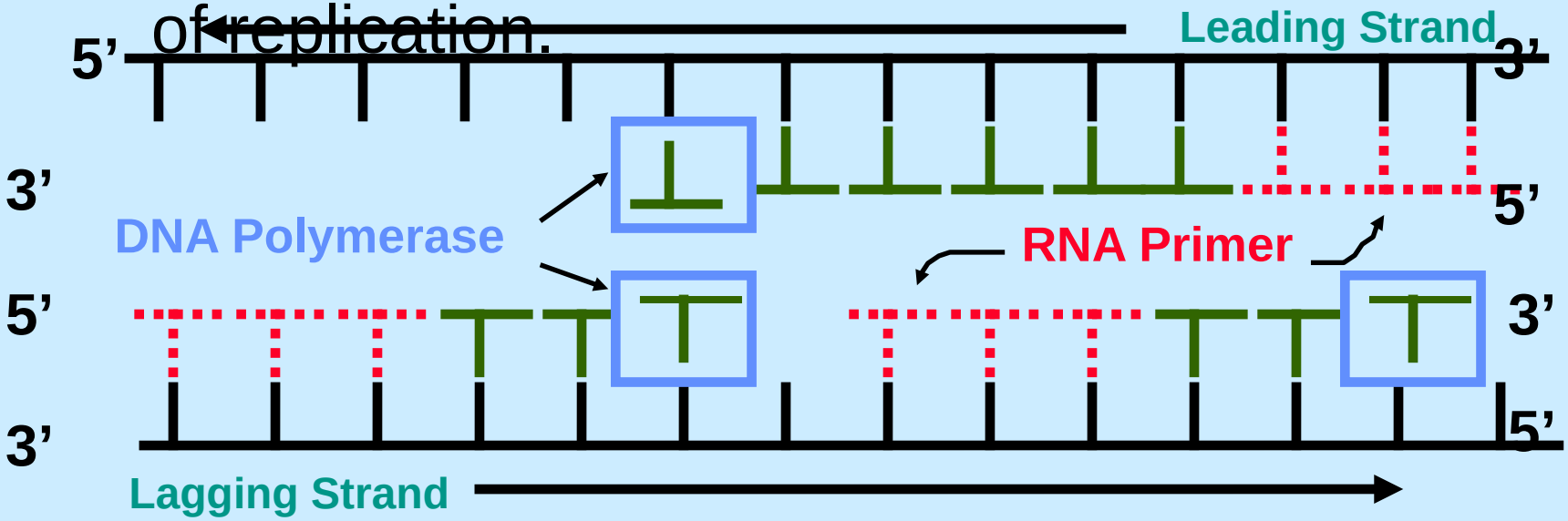
DNA Replication

2. Leading Strand: synthesized as a **single polymer** in the **5' to 3' direction**.



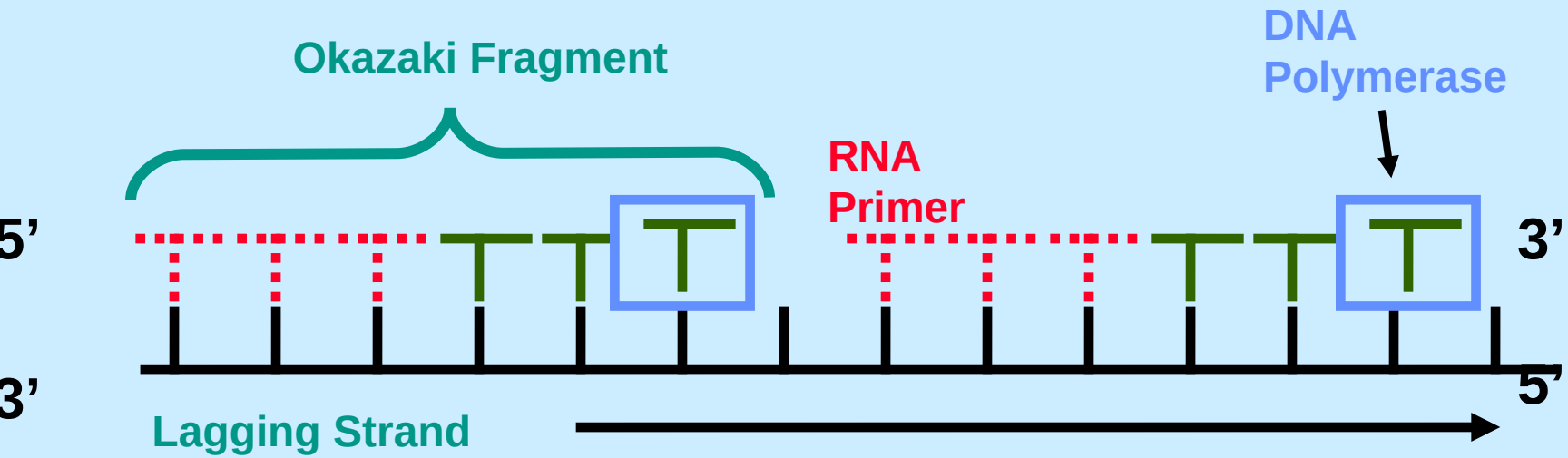
DNA Replication

3. **Lagging Strand**: also synthesized in the **5’ to 3’ direction**, but **discontinuously** against overall direction of replication.



DNA Replication

4. **Okazaki Fragments**: series of short segments on the **lagging strand.**



Primer removal and nick ligation

- There are two problems after leading and lagging strand synthesis: RNA remains in the duplex and there are nicks between each Okazaki fragment in the lagging duplex.

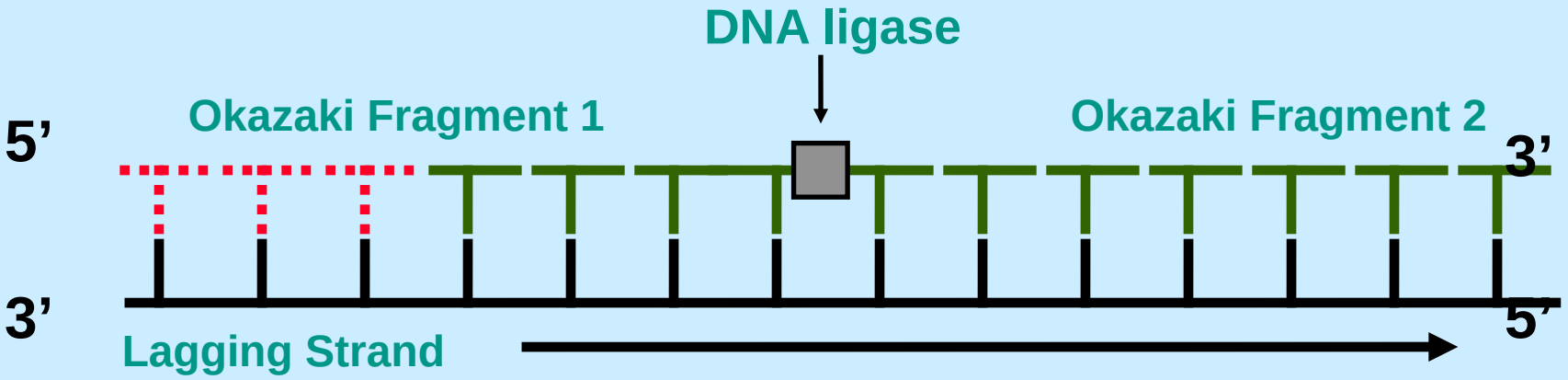
- These problems are solved by a variety of DNA repair enzymes that vary by organism, including: DNA polymerase I, DNA polymerase beta, RNAse H, ligase, and DNA2.

- This process is well-characterised in prokaryotes and much less well-characterised in many eukaryotes.

DNA Replication

5. DNA ligase: a linking enzyme that catalyzes the formation of a covalent bond from the 3' to 5' end of joining stands.

Example: joining two Okazaki fragments together.



Question:

In what direction is the newly synthesized DNA produced?

1. 5'-3'

2. 3'-5'

3. In the direction of the major groove

4. Both 5'-3' and 3'-5' depending on which strand is being replicated

Question:

- **What would be the complementary DNA strand for the following DNA sequence?**

DNA 5’-GCGTATG-3’

Answer:

DNA 5’-GCGTATG-3’

DNA 3’-CGCATAC-5’

Eukaryotic chromosomes have multiple origins of replication

1. True
2. False

What is the sequence (1 to 6) in which these proteins function during DNA replication

- ____ RNA primase
- ____ DNA ligase
- ____ DNA polymerase
- ____ Topoisomerase
- ____ DNA helicase
- ____ tethering proteins

ANSWER

What is the sequence (1 to 6) in which these proteins function during DNA replication

- __3__ RNA primase
- __6__ DNA ligase
- __4__ DNA polymerase
- __1__ Topoisomerase
- __2__ DNA helicase
- __5__ tethering proteins

Why is an RNA primer necessary for DNA replication?

A. The RNA primer is necessary for the activity of DNA ligase.

B. The RNA primer creates the 5′ and 3′ ends of the strand.

C. DNA polymerase can only add nucleotides to RNA molecules.

D. DNA polymerase can only add nucleotides to an existing strand

THANKS